Sector-wide learning does not guarantee durable returns for early green hydrogen assets

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Abstract

Low-return criteria expand modelled green-hydrogen investment eligibility, but sector-wide learning need not benefit assets built before improvements materialise. We couple 10,214 Chinese wind and utility-scale photovoltaic records to reconstructed hourly constrained electricity, installed-system costs and equity cash flows. Across six weather years, this resource is 6% of the inventory's full-output upper bound. Under the reference reconstruction, a firm rule anchored to the five-year government-bond yield admits 2,093 records; 912 fail a 6.5% comparator. Unobserved within-province allocation changes cohort sizes but not the absence of durable upgrades at terminal prices of 22 $\mathrm{CNY\,kg^{-1}}$ or less. Across source-grounded component decompositions and three conditional learning paths, stacks embody 11–42% of 2060 new-build capital savings. Crediting all contemporaneously available non-stack savings to incumbents without tax or retrofit cost upgrades at most 116 of 912 under a flat price and none with positive cost-to-price pass-through. Forward screening and pre-investment flexibility reduce exposure.

Keywords: green hydrogen, low-opportunity-cost electricity, project finance, learning curves, vintage capital, investment flexibility

Deep decarbonisation of steel, chemicals and other hard-to-abate sectors requires green hydrogen[1–3]. Global hydrogen demand surpassed 100 Mt in 2025, yet low-emissions production remained near 1 Mt; the announced 2030 pipeline contracted to

27 Mt yr^{-1} while only 4.3 Mt yr^{-1} was committed[4]. China accounts for more than 60% of committed electrolysis capacity by 2026 amid expanding wind, photovoltaic and industrial hydrogen applications[4–8]. Return criteria determine which pipeline projects reach investment. Two investment rules disclosed by state-controlled Chinese energy companies provide firm-level benchmarks. One ties the equity-return criterion for strategic emerging projects to the contemporaneous five-year government bond yield, approximately 1.45% over the evaluation window[9, 10]; the other sets a 6.5% after-tax criterion for hydrogen-based energy[11]. We use the former as a lower entry screen and the latter as a durability comparator; neither is represented as a national policy rule.

The standard argument for bridging entry and durability invokes learning. Experience curves link cumulative production or deployment to declining unit cost[12–14]. Empirical and prospective studies quantify how manufacturing, system integration, electricity use, stack life and replacement cost may improve with experience[15–26]. Deployment models consequently represent learning as a sector-level cost trajectory that shifts downwards with scale[27, 28]. Under this logic, continued deployment makes electrolysis cheaper and can eventually lift low-return early projects across a durable-return threshold.

Yet electrolyser learning and cost decline have coincided with a wide gap between announced and committed production[29]. The missing dimension is ownership of learning gains across capital vintages. Manufacturing, balance-of-plant and engineering improvements lower future-equipment cost; operating assets capture only gains embodied in replacement stacks and practice. Retrospectively crediting industry-wide decline to a commissioned plant therefore assigns it improvements that it cannot physically access. Renewable-hydrogen economics adds a second channel because conversion cost, electricity conditions and output value jointly determine returns[4, 30–32]. We call the separation the *vintage-learning paradox*: sector-wide learning can succeed while marginal assets admitted under a lower criterion remain unable to reach 6.5%.

We test this paradox by combining 10,214 operating wind and utility-scale photovoltaic records with hourly low-opportunity-cost electricity, electrolyser dispatch and sizing, nominal equity cash flow and vintage-specific learning. The framework assigns gains to the capital vintage able to capture them. Green hydrogen is a demanding test because its replaceable stack can carry later gains in cost, efficiency and lifetime into an incumbent plant[23, 25, 26]. Yet central operating learning closes only 2 of 912 strict-marginal return gaps. Across source-grounded component decompositions and three conditional learning paths, stacks embody 11–42% of 2060 new-build capital savings; even transferring all contemporaneously available non-stack savings to incumbents leaves at least 796 records unresolved under a flat price. Replaceability attenuates but does not overturn vintage lock-in. Sector-wide learning lowers future-plant entry cost while price convergence can erode incumbent revenue, exposing early capital to a structural mismatch between sector-level learning and asset-level finance.

Low-opportunity-cost electricity, rather than total renewable output, limits developable supply

System-constrained electricity compresses the physical hydrogen boundary within the covered inventory to about 6% of the full-output upper bound. The inventory contains 4,149 operating wind records and 6,065 operating utility-scale photovoltaic records, totalling approximately 629 GW: 72% of June 2024 official all-wind-plus-centralized-photovoltaic capacity, or 53% when distributed photovoltaics are included. Under the reference resource profile, total modelled renewable output is about 1,109 TWh, whereas calibrated system-constrained electricity is 70.8 TWh. At 55 $\mathrm{kWh\,kg^{-1}\,H_2}$, the full-output and constrained-electricity branches correspond to 20.2 and 1.29 $\mathrm{Mt\,H_2\,yr^{-1}}$, respectively. This bounds physical diversion and low-opportunity-cost supply within the inventory.

Temporal concentration further reduces operable scale, consistent with China's system-specific renewable curtailment[33]. Under the reference daily peak-shaving reconstruction, the median record has positive constrained electricity in approximately 1,300 hours, and the 10% highest-output constrained hours account for about 91% of annual constrained energy (Fig. 1b). Replaying six ERA5 weather years gives inventory-level hydrogen potentials of 1.28–1.34 $\mathrm{Mt\,yr^{-1}}$ and pairwise Spearman correlations above 0.996 for record-level resource rankings (Fig. 1d,e). Annual peak-shaving and proportional allocation, each conserving the same annual constrained energy, span median active durations of approximately 460 and 5,180 hours. Public monthly totals anchor seasonal magnitude; three energy-conserving reconstructions span unidentified hourly allocation[34].

Maximising nominal energy capture can over-size electrolysers. With a 30% minimum-load constraint for alkaline electrolysis, nominal capture targets of 60% and 90% require approximately 31.7 and 64.3 GW of electrolyser capacity and produce 0.75 and 1.08 $\mathrm{Mt\,H_2\,yr^{-1}}$. Extending the target to 100% raises capacity to about 163 GW, but many hours fall below the operating threshold and realised production declines to 0.78 $\mathrm{Mt\,yr^{-1}}$. Physical energy availability therefore does not translate proportionally into operable hydrogen output (Fig. 1c).

Economic entry retains only part of this physical boundary. Under the central technical and financial specification, 2,093 records meet the low-return criterion, installing 9.8 GW of electrolysers and producing 0.36 $\mathrm{Mt\,H_2\,yr^{-1}}$, about 28% of the constrained-electricity physical potential. Provincial physical potential is concentrated in Inner Mongolia, Xinjiang, Hebei, Gansu and Qinghai, but entry does not scale proportionally with potential (Fig. 1a). Holding annual constrained energy fixed while changing only its hourly allocation moves admitted hydrogen supply from 0.14 to 0.84 $\mathrm{Mt\,yr^{-1}}$. Annual resource totals bound the physical opportunity, but hourly concentration, minimum load and cash-flow conditions determine developable supply.

A lower return criterion creates a broad but heterogeneous marginal-entry band

The low-return and 6.5% criteria lie on one continuous admission frontier but separate 912 strict-marginal records. At a 2026 producer-side evaluation price of 28 CNY kg^{-1} and the central installed-system cost, the low-return criterion admits 2,093 records. When every record is independently resized to maximise net present value (NPV) at 6.5%, 1,181 remain feasible. We define the remaining 912 records, 44% of the low-return cohort, as *strict-marginal*: at least one capacity is feasible under the low criterion, but no evaluated capacity is feasible at 6.5%. As the criterion increases continuously from 1% to 8%, feasible records decline from 2,324 to 1,074; 2%, 4% and 6% retain 1,875, 1,432 and 1,215 records, respectively (Fig. 2a,b). The marginal cohort is therefore not an artefact of comparing only two discrete thresholds.

The lower criterion expands the number of admissible records more than it expands supply. The 2,093 admitted records represent 9.8 GW of electrolysers, 70.7 billion CNY of gross capital and 0.36 Mt H_2 yr^{-1}. Strict-marginal records represent 2.1 GW, 15.1 billion CNY and 0.062 Mt yr^{-1}, or approximately 21%, 21% and 17% of the respective admitted totals. Xinjiang and Qinghai contain about 59% of strict-marginal records and 76% of both their capital and hydrogen output (Fig. 2e–g). The lower criterion therefore concentrates a disproportionate share of marginal capital exposure in north-western China.

Hourly allocation changes cohort size and membership without eliminating the long-run separation. Public data identify annual constrained energy by province and technology, leaving its record-level allocation partially identified. In a common 16-candidate grid, two concentration extremes that preserve every province–technology total admit 103–2,184 records and form 4–741 strict-marginal records, compared with 1,889 and 741 under uniform within-group rates; no resulting strict cohort reaches 6.5% at terminal prices of 22 CNY kg^{-1} or less. Separately, daily peak-shaving, annual peak-shaving and proportional hourly reconstructions admit 2,093, 653 and 5,679 records and generate 912, 433 and 436 strict-marginal records. Six ERA5 weather replays admit 2,315–2,682 records and form 1,080–1,321 strict-marginal records (Fig. 2d). Cohort size varies, but no reconstruction or weather test produces durable upgrading. Here and below, durable 6.5% means that locked 2026 capacity has non-negative full-horizon equity NPV at both criteria.

Entry-price and accounting sensitivities change the wedge materially but do not create it. At evaluation prices of 18, 22, 28 and 32 CNY kg^{-1}, low-return entry is 95, 951, 2,093 and 3,494 records, and strict-marginal cohorts contain 93, 657, 912 and 1,678 records (Fig. 2c). In the coarser G16 audit, changing common nominal escalation from 0 to 3% moves low-return entry from 1,292 to 2,435 records and the strict cohort from 401 to 1,181. Relative to a 256-candidate grid, the 128-candidate grid has Jaccard similarities of 99.4% for entry and 98.2% for strict membership. Adding the exact 1-MW boundary and continuously searching 1,007 financial-boundary records changes only six identities in each cohort (Extended Data Fig. 1e).

Operating learning available to existing assets is too small to close the return gap

The initial return gap of strict-marginal assets is much larger than the operating learning they can capture. Across 912 records, the median NPV shortfall required to reach 6.5% is 22.9% of initial capital. With hydrogen held at 28 CNY kg^{-1}, median NPV gain from the central operating-learning path is 1.33% of initial capital, or 5.9% of the initial gap; only two records close it (Fig. 3a,b). A joint optimistic path closes three.

The learning-rate boundary is far outside the evidence-supported range. To avoid making this conclusion depend on an imposed replacement-cost floor, we removed that floor and scanned stack-cost learning continuously from 0 to 90% per cumulative-capacity doubling. With the central lifetime path, learning rates of 8, 13 and 18% close two, two and three records, respectively; even 90% closes only three. Fixing replacement at an extreme 20,000-h cadence raises the corresponding counts at 13, 18 and 90% learning to 70, 83 and 105, but still leaves 807 of 912 records unresolved. At 60,000 h no tested rate closes more than three, and at 80,000–100,000 h none does. Expanding replacement expenditure from the source-central 11% to a 45%-of-CAPEX component-scope bound also closes at most three records at 60,000 h across 8–50% learning. The conclusion therefore has a measured flip boundary rather than following mechanically from the definition of a strict-marginal record (Fig. 3f).

Most new-build capital learning is physically inaccessible to an incumbent. We separate the 11% stack-replacement expenditure used in cash flow from the stack's component share in installed capital. The central decomposition assigns 67.5% of installed cost to direct CAPEX and 25% of direct CAPEX to the stack, under which the replaceable stack embodies 18.2% of the capital saving achieved by a 2060 new build. Crossing direct-cost shares of 50–70%, stack shares of 20–50% of direct cost and three conditional learning paths expands this fraction to 11.3–42.3%. We then imposed a deliberately favourable upper bound: at first stack replacement, a share λ of the non-stack saving available in that replacement year is paid to the incumbent without tax or retrofit cost. In the central decomposition with price fixed at 28 CNY kg^{-1}, transferring 25, 50, 75 and 100% upgrades 3, 21, 47 and 86 records, respectively. Across the full joint boundary, complete transfer upgrades 13–116 records and therefore leaves at least 796 unresolved. At any positive pass-through β, which couples new-build cost decline to producer price, no tested combination produces a durable upgrade. Price decline therefore intensifies, but is not required for, the separation (Fig. 3e).

The terminal price and its timing determine whether a strict-marginal asset continues to meet the low-return criterion, but do not upgrade the cohort to 6.5%. With a 2060 endpoint of 22 CNY kg^{-1}, front-loaded, linear and back-loaded paths leave 8, 97 and 302 strict-marginal records above the low-return criterion; none reaches durable 6.5%. At 18 CNY kg^{-1}, only the back-loaded path preserves 80 low-return records. Solving for the terminal price required for each record to reach durable 6.5% under a linear path gives a median of 38.4 CNY kg^{-1} and a 5th–95th percentile range of 30.8–42.8 CNY kg^{-1} (Fig. 3d), quantifying how far the cohort lies from durable return.

Price decline removes substantially more value than operating learning restores. Under the linear 18 CNY kg^{-1} path, price change lowers strict-marginal NPV, discounted at the low criterion, by approximately 9.5 billion CNY; operating learning adds 0.55 billion CNY and offsets 5.8% (Fig. 3c). At 22 CNY kg^{-1}, the offset is about 9.8%. Allowing annual low-cost electricity to rise by 25% by 2060 upgrades none; other learning-anchor and resource-access tests preserve the separation (Supplementary Information). Locked assets capture replacement and operating gains, whereas manufacturing, balance-of-plant and engineering gains accrue to future plants.

Forward screening and pre-investment flexibility move risk ahead of capital lock-in

Embedding a long-run price boundary at FID reduces capital that later misses the durable-return criterion, consistent with real-options reasoning for irreversible investment[35, 36]. If low-return entry is locked and price declines linearly to 18 CNY kg^{-1}, only 280 of 2,093 records remain durable; 51.7 of 70.7 billion CNY falls below 6.5%, and durable output is 0.116 Mt H_2 yr^{-1}. A static 6.5% entry rule still leaves 11.7 of 37.4 billion CNY below the long-run criterion. By contrast, a path-informed screen retains 832 records, 20.4 billion CNY and 0.132 Mt yr^{-1} (Fig. 4a,b): 71% less committed capital but 14% more durable output than low-return locked entry. Timing-robust screening retains 476 records, 10.4 billion CNY and 0.072 Mt yr^{-1}. Zero non-durable exposure is definitional for these screens; retained capital and supply are the informative outcomes.

Forward screening prevents fragile entry but does not repair the strict-marginal cohort. None of its 912 records passes across the 12–22 CNY kg^{-1} endpoints and three timings. At 22 CNY kg^{-1}, linear-path and timing-robust screens retain 1,042 and 896 records, respectively (Fig. 4f); an endpoint-and-timing minimax screen retains only 20, none strict-marginal. Screening identifies economic headroom rather than relabelling marginal assets as durable.

Pre-investment flexibility converts resource error into cancellation and supply contraction. At 75% resource realisation, locked capacity leaves 3.8 billion CNY below the low-return criterion and produces 0.325 Mt H_2 yr^{-1}. Complete adjustment, including exit where no positive capacity passes, leaves no modelled exposure by construction: 252 records are not built, 18.5 billion CNY is not committed and output falls to 0.265 Mt. Across 50–75% realisation, it avoids 3.8–62.3 billion CNY of exposure while foregoing 0.060–0.085 Mt yr^{-1} (Fig. 4d,e,g). A separate 2025-design G16 replay across 2020–2024 weather leaves 55–406 records unbuilt, raises cohort NPV by 1.3–6.1% and changes output by -9.0 to $+0.5$%. Flexibility avoids commitment; it does not rescue sunk capital.

Retrospective support requirements exceed the illustrative limits tested here. Under the linear 18 CNY kg^{-1} path, neither a 4 CNY kg^{-1} 15-year premium nor a 25% upfront capital grant upgrades any strict-marginal record; cohort medians are

10.7 CNY kg^{-1} and 52%, respectively (Fig. 4c). The magnitude of these financing gaps shifts the actionable margin from blanket rescue towards screening before commitment.

Discussion

Sector-wide learning does not guarantee durable returns to early assets. Falling green-hydrogen cost with cumulative electrolyser deployment extends the experience-curve tradition of Wright and Arrow[12, 13]. Yet these curves aggregate learning without identifying who captures it. Once an electrolysis plant is built, manufacturing, balance-of-plant and engineering advances lower future-system cost but cannot be credited retrospectively to sunk equipment; incumbents capture only improvements embodied in replacement stacks and operation. Learning therefore separates into new-build and incumbent channels that can diverge, an incidence problem within embodied technical change and vintage capital[37]. The model treats manufacturing learning and market-price convergence as independent exogenous processes and then couples them in a reduced-form pass-through boundary: the former determines which vintage captures technical savings, whereas the latter changes incumbent revenue. Green hydrogen is a demanding test because its stack is replaceable after tens of thousands of operating hours, allowing later gains in stack cost, efficiency and lifetime to enter an operating plant[23, 25, 26]. If any early energy asset could capture later learning, an electrolyser should. Yet central operating learning closes only 2 of 912 strict-marginal gaps. Across the source-grounded component and conditional-learning boundary, stacks embody 11–42% of 2060 new-build capital savings and a tax-free, retrofit-free transfer of every contemporaneously available non-stack saving closes at most 116 gaps under a flat price; any positive cost-to-price pass-through reduces that count to zero. Replaceability attenuates but does not overturn vintage lock-in. Industry cost decline and early-asset underperformance are two incidence outcomes of this process. The question shifts from whether learning occurs to which capital vintage owns its gains.

This separation redefines when public capital can intervene. Relaxing entry on the assumption that later sector learning repairs today's assets conflates new-build and incumbent channels. First, durability screening should precede final investment decision: under the linear 18 CNY kg^{-1} path, path-informed screening commits 71% less capital yet retains 14% more durable output than low-return locked entry. Second, low-cost-electricity uncertainty should be absorbed through pre-investment capacity flexibility; at 75% resource realisation, complete adjustment withholds 18.5 billion CNY and leaves 252 records unbuilt while reducing annual output by 0.060 Mt. Third, fiscal support should remain residual and directed to supply with separately demonstrated social value. Cheaper hydrogen can accelerate downstream decarbonisation, whereas guarantees shift costs to budgets, consumers or users. Intervention should therefore focus on when capital becomes irreversible and which assets merit support.

These conclusions have defined boundaries. The non-random inventory equals 72% of contemporaneous all-wind-plus-centralized-photovoltaic capacity, or 53% including distributed photovoltaics; its absolute quantities depend on record coverage, hourly reconstruction and weather year and therefore describe the inventory, not national potential. The transferable result is a low-output cohort near the cash-flow boundary whose accessible operating learning cannot close the durable-return gap. Production assumes all hydrogen is sold at the plant gate and excludes off-site transport, storage, demand and sales constraints. A uniform 0.5–2 CNY kg^{-1} plant-gate netback penalty reduces the reoptimised strict cohort from 912 to 728–407 records and materially changes membership, confirming that exact cohort size and geography are conditional. This is particularly consequential because Xinjiang and Qinghai contain most strict-marginal exposure. Direct electrical coupling is likewise a boundary: a free lossless buffer can reverse classifications, whereas a source-anchored central battery-cost case upgrades none and low-cost cases remain contingent. Water cost is included, but basin availability, abstraction permits and seasonal competition are not[38, 39]. The firm-level equity screens are evaluated without debt-service, reserve-account or default constraints and are not national policy rules; deterministic scenarios are not probabilities, and forward screening and complete pre-FID flexibility are information-rich upper bounds. These limits constrain extrapolation, not the incidence logic. The same decomposition can test other energy technologies with replaceable core components, providing a common way to identify where transition investment is most exposed to inter-vintage mismatch.

Methods

Research design and system boundary

The analysis unit is an operating wind or utility-scale photovoltaic `ObjectId` record in the project inventory. Resource reconstruction, capacity candidates, nominal equity cash flow, entry and durability are evaluated at record level before provincial and inventory-level aggregation. Separate construction phases of one named facility can appear as multiple records, so all counts are described as project records rather than distinct physical plants. The evaluation runs from 2026 to 2060, with continuing access to low-cost electricity represented through contract renewal, host repowering or replacement supply. The primary branch uses system-constrained electricity. A mutually exclusive full-output branch redirects all modelled renewable output to electrolysis and charges the provincial feed-in price as an opportunity-cost proxy, providing a physical diversion boundary.

The financial boundary includes the installed electrolysis system: electrolyser package, rectification and electrical systems, gas–liquid separation and purification, cooling, water treatment, buildings, engineering, procurement, construction and contingency. It excludes the existing renewable host asset, off-site compression and storage, hydrogen pipelines, distribution and end-use facilities. Construction period, working capital and residual value are zero in the primary boundary, and all hydrogen is sold at the producer side. Separate tests add zero to two construction years with capitalised interest and after-tax residual values of 0–20% of initial capital. These choices define the production-side investment boundary.

Project inventory and hourly low-opportunity-cost electricity

The inventory combines the June 2024 Global Wind Power Tracker and Global Solar Power Tracker releases and contains 10,214 operating records with capacity, technology, coordinates and province[40]. The 2020 hourly reference files are station-level model outputs generated within the authors' research group and previously used by Li and Zhang[41]; that article describes simulated generation records and reconstructs hourly curtailment by daily peak-shaving to provincial rates. The files contain 8,784 leap-year hours and 15,573 unique identifiers, of which all 10,214 inventory identifiers match one-to-one. We sum the grid-delivered and constrained components only to recover a pre-allocation full-output shape; neither the earlier study's projected 2030 curtailment rates nor its hourly constrained quantities enter our annual calibration. We instead calibrate annual constrained energy to 2025 province- and technology-specific utilisation rates[42]. For province p and technology s, the annual constrained share is $r_{ps} = 1 - U_{ps}$. The reference daily peak-shaving reconstruction solves a threshold θ_{id} for record i and day d:

$$r_{ps} = 1 - U_{ps}, \qquad \sum_{h=1}^{24} \max(P_{idh} - \theta_{id}, 0) = r_{p(i)s(i)} \sum_{h=1}^{24} P_{idh}, \tag{1}$$

where P_{idh} is calibrated full output. The reference reconstruction applies r_{ps} uniformly to records within each province–technology group; record-level allocation is treated as partially identified. Two concentration extremes therefore allocate the same group total greedily to records in ascending or descending installed-capacity order, capped by each record's full output. All three allocations conserve every group total exactly. The hourly proxy is $C_{idh} = \max(P_{idh} - \theta_{id}, 0)$; thresholds are solved by bisection and annual energy is rescaled exactly. Annual peak-shaving uses one threshold over the annual duration curve, whereas proportional allocation assigns the constrained share at every hour. The resulting inputs are modelled generation and allocation profiles used to test the investment mechanism.

Weather-year sensitivity uses ERA5 hourly 100-m wind components, 2-m temperature, surface pressure and surface solar radiation downwards for 2020–2025[43, 44]. Variables are bilinearly interpolated to each record. Wind output uses an air-density-corrected generic power curve, and photovoltaic output uses a temperature-corrected irradiance model. Each record is calibrated to its 2020 reference equivalent full-load hours; this scalar is fixed in later weather years so that replay changes meteorological shape while holding the 2025 utilisation-rate calibration constant.

Figure 1a uses visible national and provincial linework derived from the Ministry of Natural Resources Standard Map Service System's 2023 China boundary product GS(2023)2767. Record coordinates are transformed with a spherical Albers equal-area display fitted to that linework; the internal province GeoJSON is used only for invisible coordinate registration and thematic clipping, not as the displayed boundary source. The transformation, diagnostic overlay, source files and hashes are archived with the figure data. The modified thematic composite remains subject to final map review before publication.

Electrolyser dispatch and capacity choice

For candidate electrolyser capacity K, the plant operates only when available low-opportunity-cost power exceeds the minimum-load fraction λK. Absorbed power in hour t is

$$A_{it}(K) = \mathbb{1}(C_{it} \geq \lambda K) \min(C_{it}, K). \tag{2}$$

Annual absorbed electricity, operating hours and hydrogen output are

$$\begin{aligned} E_i(K) &= \sum_t A_{it}(K)\Delta t, \\ h_i(K) &= \sum_t \mathbb{1}(A_{it} > 0)\,\Delta t, \\ H_{iy}(K) &= 1000 \sum_t \frac{A_{it}(K)\Delta t}{\varepsilon_{iyt}}. \end{aligned} \tag{3}$$

Here $\Delta t = 1\,\mathrm{h}$, E_i is measured in MWh, h_i in hours and H_{iy} in kg; the factor 1000 converts MWh to kWh. The alkaline main case uses $\lambda = 0.30$. Candidate capacities are generated from 128 nested resource-capture fractions, with the exact 1-MW engineering lower bound added as an independent candidate. Candidates below 1 MW or with zero production are removed. For return criterion r, every record is independently resized by

$$K_i^*(r) = \underset{K \in \mathcal{K}_i,\, K \geq 1\,\mathrm{MW}}{\arg\max} \mathrm{NPV}_i(K; r). \tag{4}$$

A strict-marginal record has at least one capacity with non-negative NPV at the low criterion and no capacity with non-negative NPV at 6.5%. Grid density, the 1-MW boundary, continuous local search, lower minimum loads and proton-exchange membrane (PEM) operation are assessed in the Supplementary Information. Dispatch is directly coupled without a battery or hydrogen buffer, so the minimum-load and oversizing results describe the unbuffered operating configuration.

Installed-system cost, nominal equity cash flow and return criteria

For capacity K_i in MW and installed-system unit cost c_{sys} in CNY kW^{-1}, initial investment, debt and equity are

$$I_i = 1000 K_i c_{\mathrm{sys}}, \qquad D_{i0} = \delta I_i, \qquad Eq_{i0} = (1-\delta) I_i, \tag{5}$$

where δ is the debt share. The World Bank reports 500–1,500 USD kW^{-1} for installed alkaline systems; using a transparent conversion assumption of 7.2 CNY per USD gives deterministic bounds of 3,600 and 10,800 CNY kW^{-1}, with 7,200 CNY kW^{-1} as the midpoint central case[26]. The IEA's 600–1,200 USD kW^{-1} China interval provides contextual equipment-cost evidence, while the World Bank interval defines the installed-system boundary used here[45]. Fixed operations and maintenance (O&M) is 5% of initial capital per year, and an explicit stack replacement costs 11% when triggered[25]. Mutually exclusive all-in operating-cost conventions are tested to avoid double counting[26]. Constrained electricity costs 0.10 CNY kWh^{-1} centrally and 0–0.20 CNY kWh^{-1} in sensitivity. Water uses provincial values reported in Supplementary Table 3 of the earlier study[41] and 15 kg of water per kg H_2 centrally, with 10–60 kg kg^{-1} tested as a direct-cost boundary.

All model price and cost scenarios are expressed in constant 2026 CNY and converted to nominal cash flow using a common escalation rate π. The 28 CNY kg^{-1} entry reference is anchored numerically to the December 2024 cross-route producer sample:

$$X_y^{\text{nom}} = X_y^{\text{real}}(1+\pi)^{y-2026}, \qquad R_{iy} = H_{iy}P_y(1+\pi)^{y-2026}. \tag{6}$$

The central accounting case uses $\pi = 2\%$ and tests 0%, 1% and 3%. Nominal loan rates and nominal return criteria are applied only to nominal cash flows. The central financing structure is 70% debt, a 3.5% nominal loan rate, a 15-year term and two years of principal grace. Corporate income tax is 25% and tax losses can be carried forward for five years[46]; a ten-year straight-line depreciation schedule is a transparent accounting assumption applied to initial and replacement capital.

Annual free cash flow to equity subtracts operations, stack replacement, tax, interest and principal from revenue. Classification is based on NPV consistency because replacement expenditure can create multiple cash-flow sign changes:

$$\begin{aligned} CF_{iy}^{Eq} &= R_{iy} - O_{iy} - M_{iy} - T_{iy} - Int_{iy} - Prin_{iy}, \\ \text{NPV}_i(K;r) &= -Eq_{i0} + \sum_{y=1}^{Y} \frac{CF_{iy}^{Eq}}{(1+r)^y}. \end{aligned} \tag{7}$$

A record passes criterion r when the NPV of its criterion-specific optimum is non-negative. “Durable 6.5%” means that the locked 2026 investment and capacity produce non-negative full-horizon equity NPV at both the low-return and 6.5% criteria. Capacity remains fixed after construction, and classification uses full-horizon equity value without debt-service-coverage, reserve-account or default tests.

We operationalise CHN Energy Changyuan’s 2026 rule with the mean nominal five-year ChinaBond government bond yield over 20 trading days from 4 June to 2 July 2026, reported as approximately 1.45%[9, 10]. “Five-year” denotes bond maturity, and the unrounded arithmetic mean is retained in the machine-readable calculation. The 6.5% comparator comes from Huadian Energy’s separate 2025 rule for hydrogen-based energy[11]. The two publicly disclosed firm-level criteria define the institutional comparison.

Vintage-specific operating learning and price paths

The initial alkaline-stack electricity use is $\varepsilon_0 = 55\,\text{kWh}\,\text{kg}^{-1}\,H_2$ and stack life is 60,000 operating hours. Within-stack degradation is represented by

$$\varepsilon(a) = \varepsilon_0(1 + \gamma a), \qquad \gamma = \frac{2(57.3/55 - 1)}{60000}, \tag{8}$$

so average lifetime electricity use is $57.3\,\mathrm{kWh\,kg^{-1}\,H_2}$. An operating asset receives improved efficiency, life and replacement cost only when a stack replacement is triggered by cumulative operating hours. Manufacturing learning in complete systems and balance-of-plant/engineering affects future new investment and is never used to reduce the 2026 sunk capital or annual fixed O&M of an existing plant.

The stack-cost experience curve is

$$c_{\mathrm{stack},t} = c_{\mathrm{stack},0}\left(\frac{Q_t}{Q_0}\right)^{-b}, \qquad b = \frac{-\ln(1 - LR)}{\ln 2}, \tag{9}$$

where Q_t is cumulative deployment and LR is the learning rate. The central path starts at 4 GW, a rounded end-2025 global installed-capacity anchor[4]. Author-defined conditional anchors, contextualised by published deployment ranges, then reach 140, 700, 1,400 and 2,200 GW in 2030, 2040, 2050 and 2060. A 20-GW starting normaliser provides the anchor sensitivity. The central path applies an author-selected 13% conditional experience rate to future equipment and replacement-stack cost and 5% to future balance-of-plant and engineering; applying system-level evidence to replacement stacks is the vintage-transfer assumption tested by the model. Electricity use and stack life are bounded trajectories interpolated in log-deployment space: they move from 55 to $50\,\mathrm{kWh\,kg^{-1}\,H_2}$ and from 60,000 to 90,000 h by 2060[23, 25, 26, 45]. The central replacement-stack cost factor reaches its imposed 0.55 floor in 2028 and remains there; an existing record obtains the applicable factor only in its actual replacement year.

Three incidence and falsification boundaries test whether the result follows from restricted learning assumptions. The World Bank describes direct and indirect CAPEX as broadly comparable, reports 65:35–70:30 direct-to-indirect shares for large greenfield projects and places the stack at 20–50% of direct CAPEX[26]. We therefore use $d \in \{0.50, 0.675, 0.70\}$ and $q \in \{0.20, 0.25, 0.35, 0.50\}$, with central values $d = 0.675$ and $q = 0.25$. Installed capital is decomposed into stack ($s_S = dq$), other direct equipment ($d - s_S$) and indirect balance-of-plant/engineering ($1 - d$). The central $s_S = 0.16875$ is a component-accounting share; the separate 11% replacement-event expenditure remains a cash-flow assumption. The share of new-build capital savings embodied in the replaceable stack is

$$\phi_t = \frac{s_S(1 - f_t^S)}{s_S(1 - f_t^S) + (d - s_S)(1 - f_t^E) + (1 - d)(1 - f_t^B)}, \tag{10}$$

where f_t^S, f_t^E and f_t^B are the stack, other direct-equipment and indirect-capital cost factors. Under the central path, $\phi_{2060} = 0.182$; the full component-share and learning-path boundary is 0.113–0.423. A deliberately favourable transfer boundary then credits an incumbent at its first stack replacement with

$$T_{it} = \lambda I_{i0}\left[(d - s_S)(1 - f_t^E) + (1 - d)(1 - f_t^B)\right], \qquad \lambda \in [0, 1], \tag{11}$$

where I_{i0} is initial installed-system CAPEX and λ is evaluated in 0.01 increments. The transfer is added directly to equity cash flow without tax, retrofit cost or repetition, making it an upper bound rather than a proposed instrument. A reduced-form pass-through elasticity links future new-build cost to producer price,

$$I_t^{\text{new}} = df_t^E + (1-d)f_t^B, \qquad P_t = 28\,(I_t^{\text{new}})^{\beta}, \qquad \beta \in \{0, 0.25, 0.50, 0.75, 1\}. \quad (12)$$

This is an incidence boundary, not an estimated price elasticity or a supply–demand forecast. An auxiliary test separately scales incumbent access to electricity-use, stack-life and replacement-cost improvements from zero to the central path.

The stack-learning falsification boundary removes the 0.55 replacement-cost floor and scans the stack-cost learning rate from 0 to 90% per doubling in one-percentage-point increments. It combines the central endogenous lifetime path with fixed replacement cadences of 20,000, 40,000, 60,000, 80,000 and 100,000 operating hours. The event replacement expenditure is separately varied from 6 to 45% of installed capital. These tests ask how extreme learning strength, replacement frequency and replacement scope must become before the strict-marginal conclusion reverses; they are not forecasts of rational replacement policy.

Long-run hydrogen prices are conditional paths. For terminal price P_{2060} and timing function g_s, the constant-2026-CNY price is

$$P_y = P_{2026} + (P_{2060} - P_{2026})g_s(\tau_y), \qquad \tau_y = \frac{y - 2026}{2060 - 2026}, \qquad g_s \in \{\sqrt{\tau}, \tau, \tau^2\}, \quad (13)$$

where the three functions denote front-loaded, linear and back-loaded convergence. The initial price is 28 CNY kg^{-1} and terminal values are 22, 18, 15 and 12 CNY kg^{-1}. Paired counterfactuals identify price and operating-learning contributions:

$$\begin{aligned} \Delta\text{NPV}_{\text{price}} &= \text{NPV}(P_s, L_0) - \text{NPV}(P_0, L_0), \\ \Delta\text{NPV}_{\text{learn}} &= \text{NPV}(P_s, L) - \text{NPV}(P_s, L_0). \end{aligned} \quad (14)$$

The reference lifetime calculation holds low-cost electricity equivalent hours at their 2025-calibrated level. A persistence audit scales annual active hours, absorbed electricity and electricity expenditure by a linear factor from 1 in 2026 to 0.50, 0.75, 1.00 or 1.25 in 2060, spanning declining and expanding contract-equivalent access.

Forward screening, capacity flexibility and support boundaries

Forward screening places a long-run price endpoint and timing in the capacity-choice problem before FID and retains only capacities with non-negative NPV at both criteria. Timing-robust screening first selects the engineering-eligible capacity with the greatest worst-case 6.5% NPV across $\mathcal{S}$, the front-loaded, linear and back-loaded paths:

$$K_i^{\mathrm{rob}} = \operatorname*{arg\,max}_{K \in \mathcal{K}_i^{\mathrm{elig}}} \min_{s \in \mathcal{S}} \mathrm{NPV}_i(K; 6.5\%, s), \tag{15}$$

$$S_i^{\mathrm{rob}} = \mathbb{1}\left[\min_{s \in \mathcal{S}} \mathrm{NPV}_i(K_i^{\mathrm{rob}}; 6.5\%, s) \geq 0\right] \mathbb{1}\left[\min_{s \in \mathcal{S}} \mathrm{NPV}_i(K_i^{\mathrm{rob}}; r_{\mathrm{low}}, s) \geq 0\right]. \tag{16}$$

Here $\mathcal{K}_i^{\mathrm{elig}}$ is the engineering-eligible capacity set. A selected record is retained only when $S_i^{\mathrm{rob}} = 1$, matching the implemented sequence of capacity selection followed by the two-criterion worst-case screen. A stricter minimax screen replaces $\mathcal{S}$ with the Cartesian product of the three timing shapes and terminal prices from 12 to 22 CNY kg^{-1} in 1-CNY increments. Because project NPV is monotone in price, this endpoint-and-timing screen is governed by the 12 CNY kg^{-1} boundary but is reported separately to make the information assumption explicit.

Capacity flexibility describes the share that can be adjusted before construction after a resource deviation is known. If K_i^0 is original capacity and $K_i^*(\rho)$ is the re-optimised capacity under resource realisation ρ, installed capacity at adjustability a is

$$K_i(a, \rho) = K_i^0 + a\left[K_i^*(\rho) - K_i^0\right], \qquad a \in \{0, 0.25, 0.50, 0.75, 1\}. \tag{17}$$

Among positive engineering-eligible capacities, $K_i^*(\rho)$ maximises low-criterion NPV. If none has non-negative NPV, $K_i^*(\rho) = 0$, representing cancellation before commitment; interpolated capacities below 1 MW are also mapped to zero. Exposed capital is gross initial investment in positive-capacity records that cease to satisfy the low-return criterion after the resource deviation. Avoided capital is planned investment withheld through downsizing or cancellation. Complete flexibility assumes that resource error is resolved before FID and that capacity can be adjusted without cancellation, lead-time or interconnection costs. Support boundaries search for the minimum constant 2026–2040 producer-price premium or upfront capital grant that makes a locked strict-marginal record satisfy 6.5%, yielding equivalent financing gaps for comparison with forward screening.

Robustness, uncertainty and quality control

Deterministic robustness spans installed-system capital cost, constrained-electricity price, operating cost, resource realisation and persistence, debt share, nominal loan rate, escalation, evaluation horizon, minimum load, alkaline and PEM technology, learning intensity, learning-start anchor, terminal hydrogen price, construction duration, residual value, producer-side netback and electrical buffering. The constrained-electricity branch contains 972 factorial combinations, and a mutually exclusive full-output branch adds 324 opportunity-cost combinations, for 1,296 deterministic cases in total. All combinations have equal diagnostic status and define a robustness domain. Within each province–technology group, two annual-energy-conserving concentration extremes assign constrained energy first to smaller or larger records, respectively, partially identifying outcomes under the unknown allocation. The financing contribution to within-grid dispersion is evaluated over its specified parameter range. Netback penalties of 0–4 CNY kg^{-1} are applied before reoptimising entry. Electrical buffers of zero to four equivalent hours are dispatched against the original hourly profiles; free lossless storage defines a physical upper bound, whereas costed cases include initial capital, fixed operation and maintenance, round-trip loss and periodic replacement.

Capacity discretisation is checked against 128- and 256-candidate nested grids and continuous local search. Weather shape is checked by replaying ERA5 2020–2025. In addition to the M129 resource check, the complete entry, learning and support calculations are replayed with a common 16-candidate grid for each weather year; these G16 counts are reported as a separate robustness experiment and are never pooled with the M129 main cohort. Set stability is measured by

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{18}$$

All figures and reported values are generated from record-level output tables. Automated checks cover annual energy conservation, cohort denominators, criterion nesting, nominal cash-flow reconstruction, capacity-search convergence and figure–text consistency. Provincial water price and constrained-electricity settlement are retained as transparent boundaries where evidence is insufficient for causal or implementation claims.

Data availability

Redistributable project inventory fields, annual utilisation data, processed ERA5 outputs, parameter-source registers and figure source data will be deposited in a public repository for peer review and publication [repository and DOI to be inserted before submission]. The source ERA5 variables are publicly available from the Copernicus Climate Data Store under its applicable terms. The laboratory-generated 2020 monthly hourly files and the two derived 10,214-by-8,784 hourly profile arrays are held by the authors' research group and will be documented with filenames, dimensions, checksums, analytical roles, the earlier article and its public code commit. These files are available from the corresponding author upon reasonable request for research use, subject to contributor and institutional approval and applicable data-sharing conditions. They are model outputs and are not represented as independently observed station dispatch.

Code availability

Code for constrained-electricity reconstruction, ERA5 replay, electrolyser dispatch and capacity optimisation, nominal equity cash flow, learning-flip boundaries, forward screening, capacity flexibility and figure generation will be deposited in a public repository for peer review and publication [repository and DOI to be inserted before submission]. The archive will include an environment specification, execution order and SHA-256 hashes for inputs and outputs.

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Author contributions

Jinwei Liang: Conceptualization, Methodology, Visualization, Writing – original draft. Haoran Zhang: Conceptualization, Methodology, Writing – review & editing, Supervision.

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Competing interests

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Additional information

Supplementary information accompanies this manuscript. Correspondence and requests for materials should be addressed to Haoran Zhang (h.zhang@pku.edu.cn).

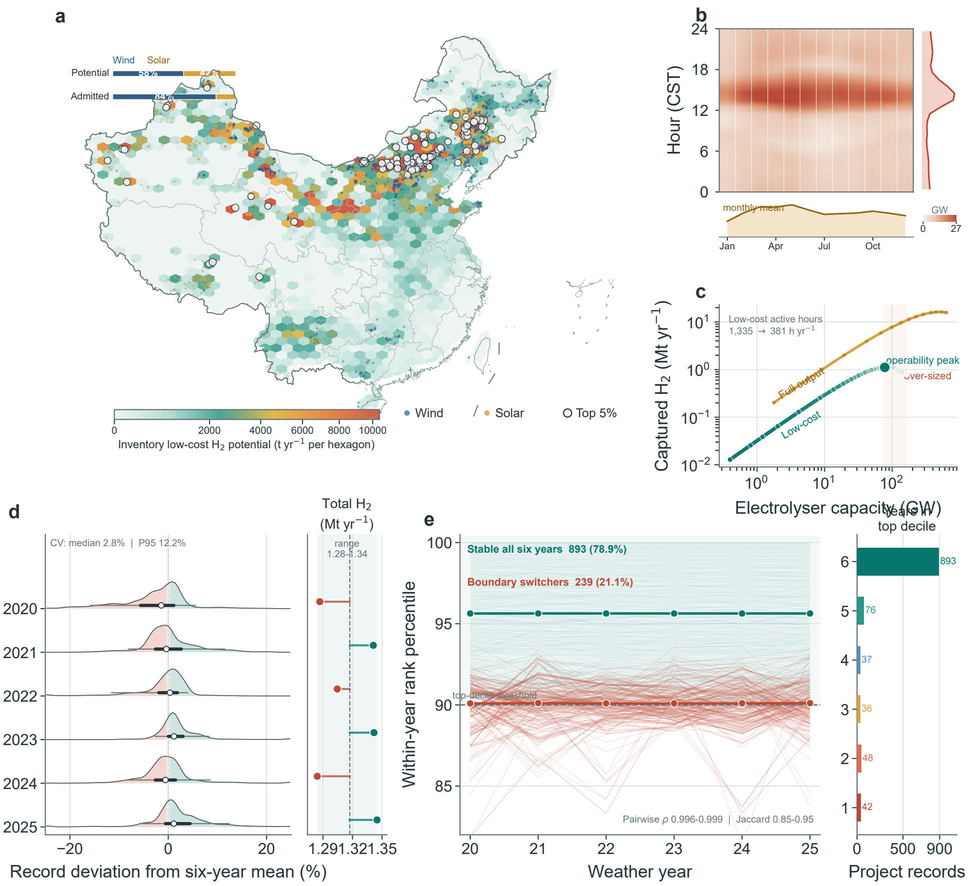


Fig. 1 Spatial, temporal and operating constraints delimit hydrogen production from low-opportunity-cost electricity within the project-record inventory. a, Spatial distribution of hydrogen potential from system-constrained electricity; dark records meet the low-return criterion and open circles identify the top 5% of admitted supply. Visible administrative linework and the South China Sea representation are derived from the Ministry of Natural Resources Standard Map Service System's 2023 China boundary map. **b**, Month–hour climatology with diurnal and seasonal margins. **c**, Operating frontier for electrolyser capacity, realised hydrogen and active low-cost hours under the mutually exclusive constrained-electricity and full-output branches. **d**, Record-level deviations from six-year mean hydrogen potential and inventory totals for ERA5 weather years 2020–2025; white points, thick lines and thin lines denote medians, interquartile ranges and 5th–95th percentiles. **e**, Rank trajectories and persistence among records in the top resource decile.

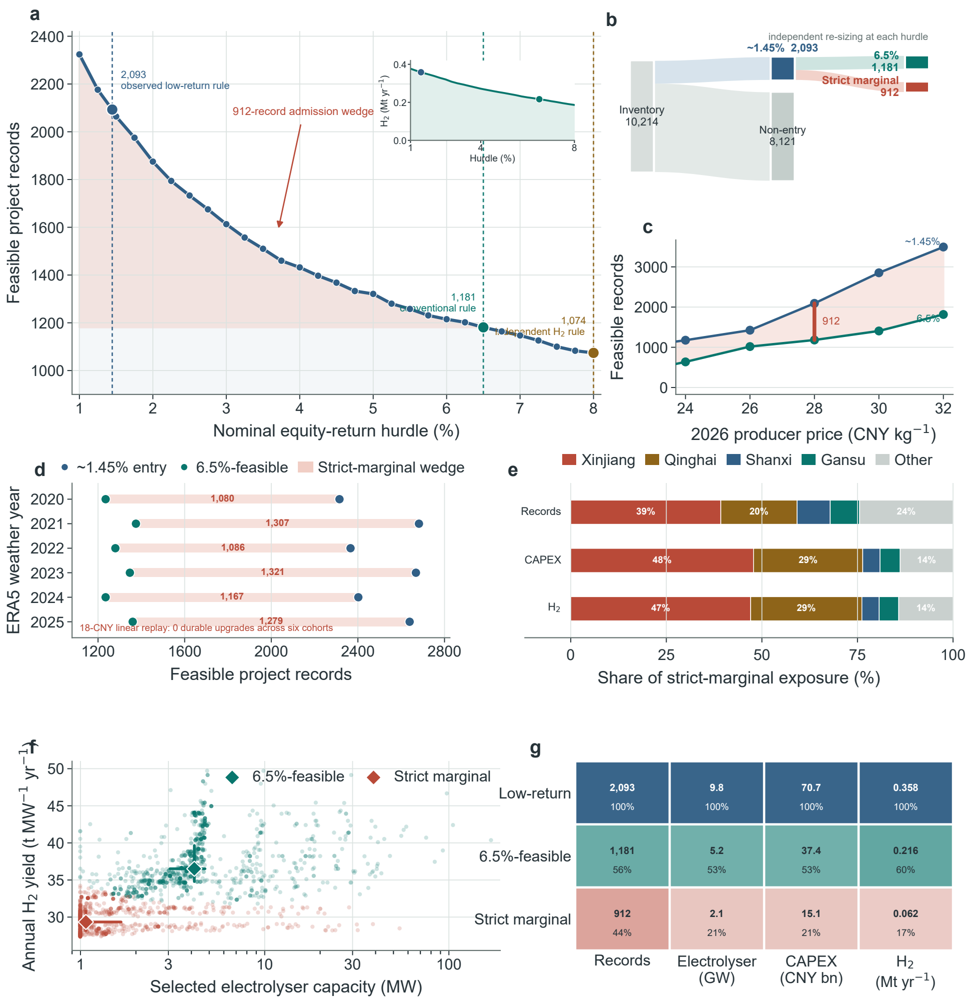


Fig. 2 Observable low-return and 6.5% enterprise criteria form a continuous and spatially heterogeneous admission boundary. **a**, Feasible project records and annual hydrogen output after independent resizing across 1–8% equity-return criteria; shading denotes records admitted relative to 6.5%. **b**, Flow from the 10,214-record inventory to low-return entry, non-entry, 6.5%-feasible and strict-marginal cohorts. **c**, Sensitivity of both criteria and their admission wedge to the 2026 producer-side evaluation price. **d**, Full-chain replay under ERA5 weather shapes for 2020–2025 using a common capacity grid; endpoints mark low-return entry and 6.5%-feasible records, and the connecting band is the strict-marginal cohort. The annotation reports durable upgrades under the linear 18 CNY kg^{-1} path. **e**, Provincial shares of strict-marginal records, gross capital and annual hydrogen output. **f**, Selected electrolyser capacity and annual hydrogen yield within the low-return cohort; diamonds and orthogonal bars denote medians and interquartile ranges. **g**, Absolute record, capacity, capital and hydrogen scales of the three entry cohorts; percentages are relative to the low-return cohort within each metric.

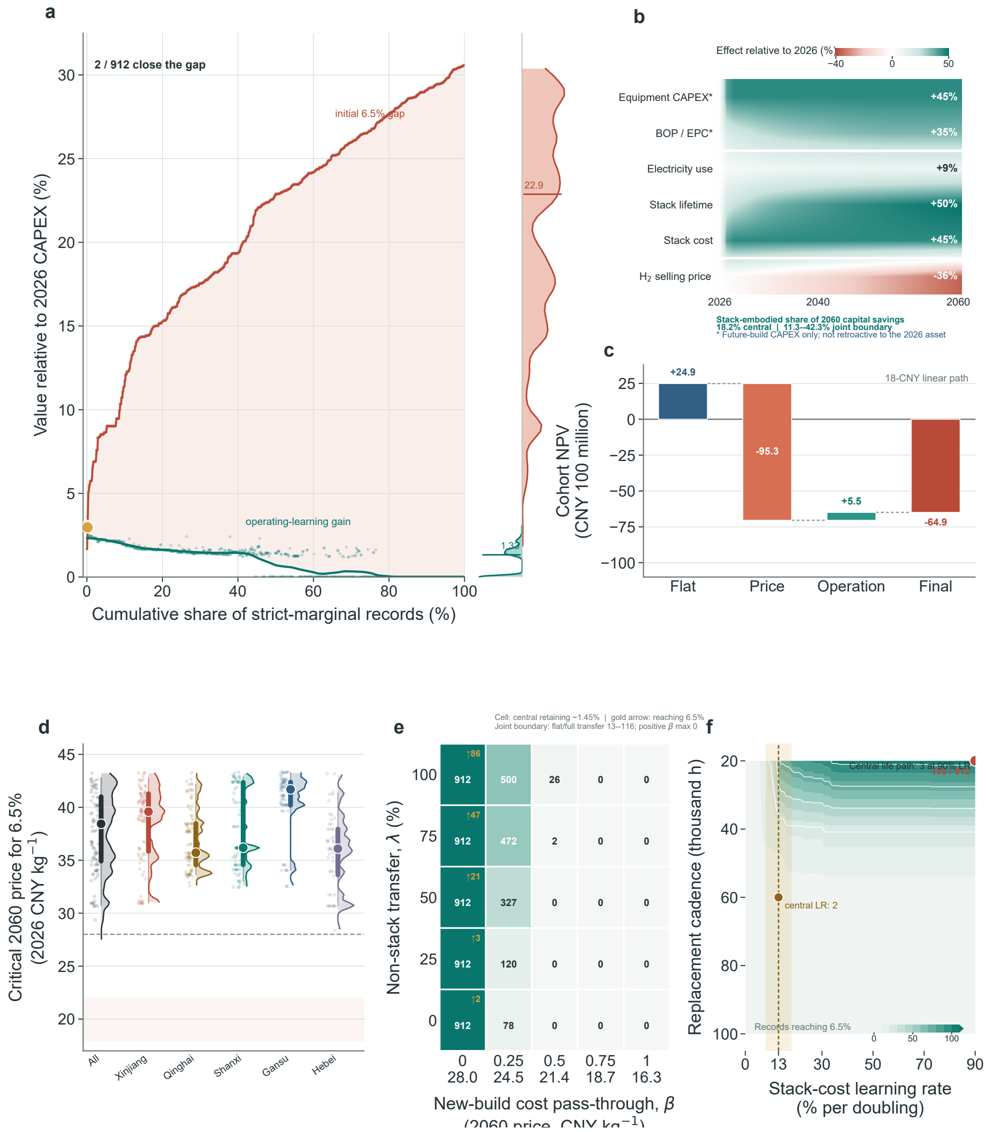


Fig. 3 Vintage-specific operating learning is small relative to the return gap of strict-marginal assets. **a**, Initial NPV gap to 6.5% versus the central operating-learning gain for 912 strict-marginal records, with marginal distributions. **b**, Relative evolution of learning and price variables from 2026 to 2060; direct-equipment and balance-of-plant/engineering learning apply only to future assets, and the annotation gives the central and joint-boundary stack-embodied shares of 2060 new-build capital savings. **c**, Paired counterfactual decomposition of the strict-marginal cohort's low-criterion NPV under the linear 18 CNY kg^{-1} path. **d**, Distribution of the 2060 terminal price required for durable 6.5%, overall and in the principal exposed provinces. **e**, Deliberately favourable transfer of otherwise inaccessible non-stack savings to incumbents at first replacement (λ), against pass-through from future new-build installed-capital decline to producer price (β); cell values and shading give central-case records retaining the low-return criterion, gold arrows give records reaching 6.5%, and the annotation reports the joint component-share and learning-path boundary. The second line of each horizontal-axis label gives the implied 2060 price. **f**, Durable-return boundary after removing the replacement-cost floor and varying the stack-cost learning rate continuously against fixed replacement cadence; shading gives records reaching 6.5%, and the gold band spans 8–18% per doubling.

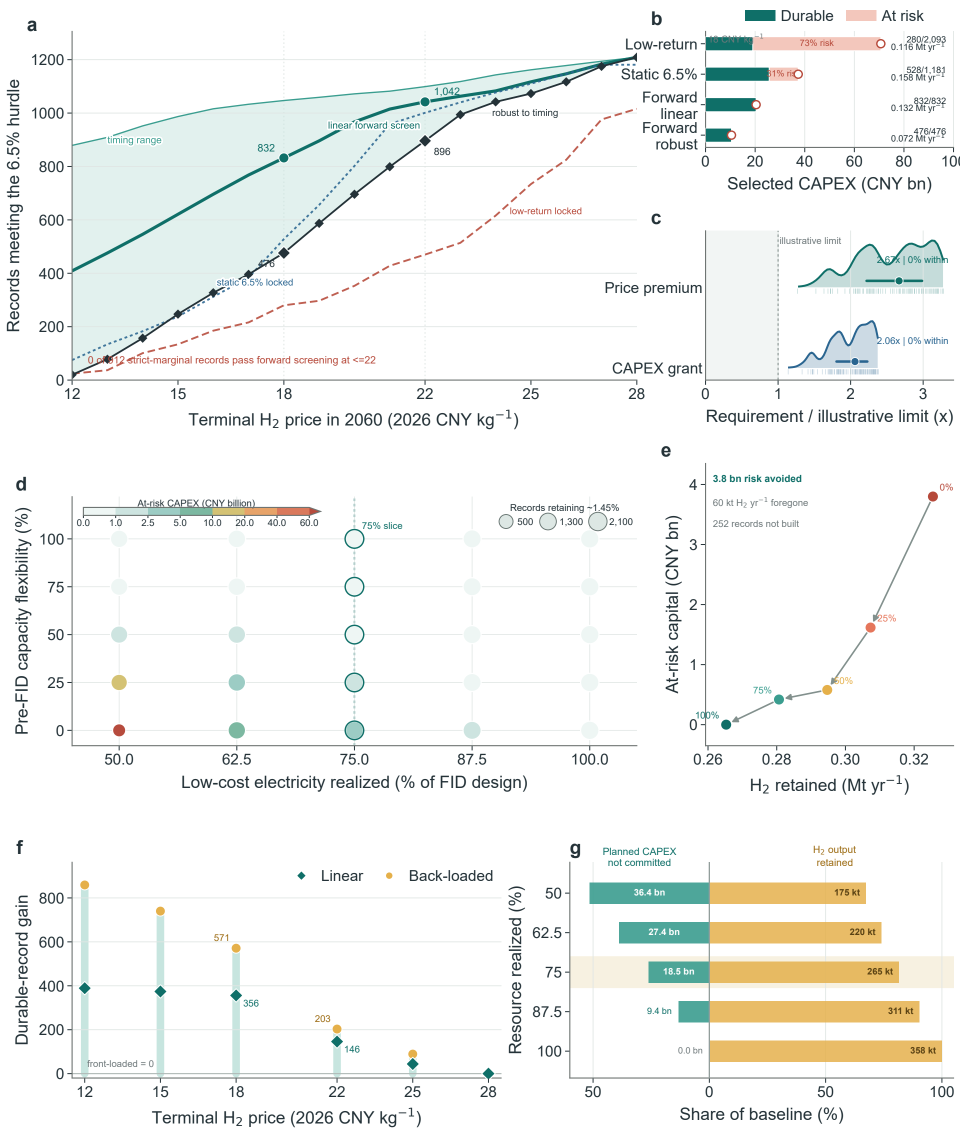
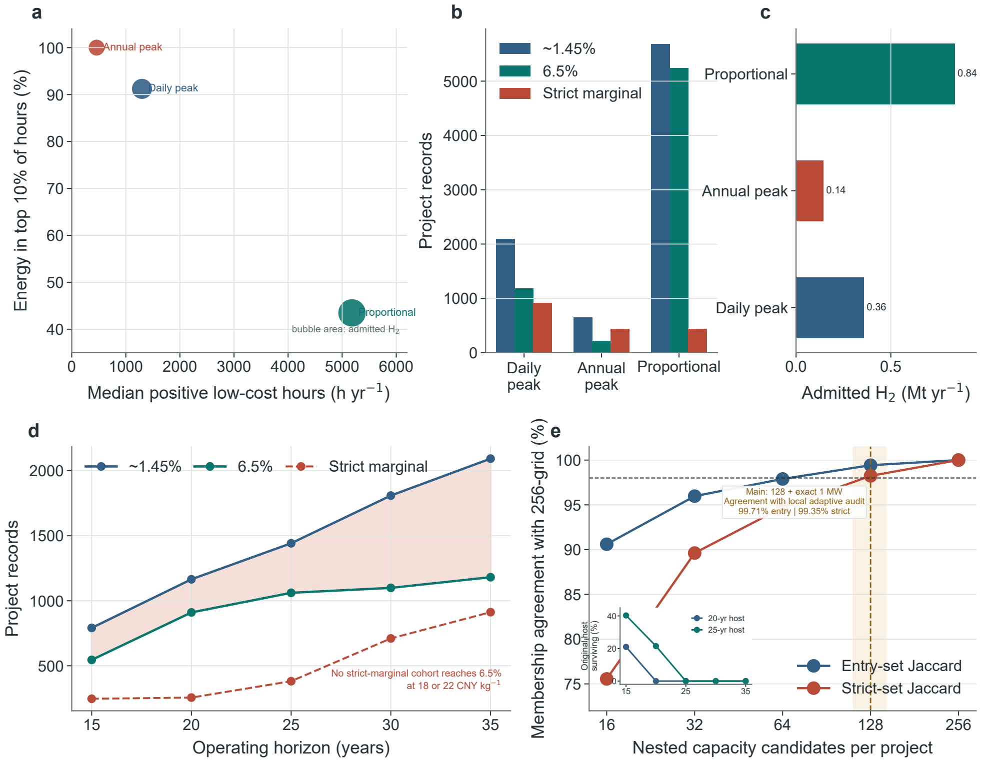


Fig. 4 Forward screening and pre-investment capacity flexibility move low-return entry risk ahead of capital lock-in. a, Durable records retained by alternative entry rules across conditional terminal prices of 12–28 CNY kg^{-1}; the green band spans front-loaded to back-loaded timing, the heavy line denotes the linear path and black diamonds denote timing-robust screening. Conditional path-specific screens assume that the future path is known at FID. **b**, Durable and at-risk capital selected by four rules for the linear 18 CNY kg^{-1} path; open circles mark total selected capital, and right-hand labels report durable/selected records and durable annual hydrogen. **c**, Required 15-year price premiums and upfront capital grants normalized by illustrative test limits; ridges show the distributions, baseline ticks sample individual requirements, thick bars show interquartile ranges, circles mark medians and shading denotes requirements at or below each limit. **d**, Discrete resource-realisation and pre-FID-flexibility sensitivity matrix; colour denotes capital below the low-return criterion and circle area denotes records retaining that criterion. **e**, At-risk capital and hydrogen output along the 75% resource-realisation slice as pre-FID flexibility rises from 0 to 100%; complete flexibility includes cancellation where no positive capacity passes the low-return criterion. **f**, Additional durable records retained by path-specific forward screening relative to timing-robust screening under linear and back-loaded price paths; front-loaded timing yields no gain because it defines the adverse timing boundary. **g**, Complete-adjustment outcomes across five resource-realisation levels; left bars show planned CAPEX not committed, and right bars show hydrogen output retained, each as a share of the locked 100%-resource baseline with absolute values annotated.



Extended Data Fig. 1 Hourly reconstruction, evaluation-horizon and capacity-search robustness. a–c, Full-chain admission and strict-marginal outcomes under the daily peak-shaving, annual peak-shaving and proportional-allocation proxies. **d**, Durable-return outcomes across 15–35-year evaluation horizons. **e**, Membership and aggregate convergence across nested capacity grids, the exact 1-MW boundary and continuous local search. Exact project-record counts remain conditional on the hourly proxy, while the failure of strict-marginal cohorts to reach durable 6.5% at terminal prices of 22 CNY kg^{-1} or less is stable across the tested boundaries.